Permutations & Combinations

Section 1: Fundamental Counting & Factorials

1 How many 4-digit numbers can be formed using the digits $1, 2, 3, 4, 5, 6$ without repetition?

a. 360

b. 720

c. 120

d. 240

2 In how many ways can 7 runners finish 1st, 2nd, and 3rd in a race, assuming no ties?

a. 210

b. 35

c. 5040

d. 120

3 How many even 3-digit numbers can be formed using digits $1, 2, 3, 4, 5$ without repetition?

a. 24

b. 12

c. 36

d. 60

Section 2: Permutations with Repetition & Restrictions

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4 How many distinct arrangements of all the letters in the word MATHEMATICS are possible?

a. $4,989,600$

b. $39,916,800$

c. $1,247,400$

d. $2,494,800$

5 In how many ways can 5 boys and 4 girls stand in a line if all 4 girls must stand together?

a. $17,280$

b. $2,880$

c. $43,200$

d. $5,760$

6 In how many ways can 6 people sit around a circular table, where rotations are considered equivalent?

a. 120

b. 720

c. 24

d. 360

Section 3: Combinations & Committee Selection

7 A committee of 5 is to be chosen from a group of 8 men and 6 women. How many committees are possible if it must contain exactly 3 men and 2 women?

a. 840

b. 420

c. 1260

d. 2002

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8 From a group of 10 students, a team of 4 is selected. If the team must include the student council president, how many distinct teams are possible?

a. 84

b. 210

c. 126

d. 72

9 A team of 4 is selected from 7 boys and 5 girls. How many teams have at least one girl?

a. 460

b. 495

c. 420

d. 35

Section 4: Complex Multi-Condition Problems

10 How many 5-digit odd numbers greater than 40,000 can be formed using digits $2, 3, 4, 5, 7$ without repetition?

a. 28

b. 36

c. 48

d. 24

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11 A delegation of 4 is chosen from 5 teachers and 6 students. In how many ways can this be done if there must be more teachers than students?

a. 75

b. 65

c. 85

d. 50

12 Four cards are drawn from a standard 52-card deck. What is the probability that all four cards are of the same suit?

a. $\frac{44}{4165} \approx 0.01056$

b. 0.0025

c. 0.0450

d. 0.0210

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